Melih Barsbey

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SUMMARY

AI researcher working on reliable and efficient machine learning. Current research focuses on safety, alignment, and post-training stability in LLMs and VLMs under resource constraints, and how representation geometry mediates the effects of these constraints. Research published in *NeurIPS*, *ICML*, *ICLR*, *ICCV*, and *Nature Medicine* among others; Area Chair for *ICLR*. Further research topics include mechanistic interpretability, learning theory, and medical decision-making with human-AI complementarity and noisy supervision. Industry experience in applied ML research, deployment, and team leadership, including daily demand forecasting across 8,000+ ATMs.

PROFESSIONAL EXPERIENCE

Imperial College London, London, United Kingdom

- Postdoctoral Research Associate in the Department of Computing. Jul 2024 – Present
  - Jailbreak and prompt injection vulnerability of LLMs and VLMs under quantization, pruning, and low-rank factorization.
  - Effects of weight, activation, and KV cache quantization on safety, alignment, and post-training stability; evaluation and mitigation strategies.
  - Additional published research on robustness to spurious correlations, mechanistic interpretability, topological deep learning for drug discovery, grokking and learning theory.

Google DeepMind, London, United Kingdom

- Research Scientist Intern, Robust and Verified AI team. Jun 2022 – Oct 2022
  - Developed a confidence-based deferral system for robust human-AI decision-making with sparse medical data.
  - Evaluated and trained models under noisy, conflicting annotations: ranking models over partially ordered expert judgments; the k-wise generalization of the Bradley-Terry preference model.

Freelance Machine Learning Consultant

- Freelance consultancy for multiple software companies on the research, development, and deployment of machine learning systems. Jul 2019 – Jun 2024
  - Development of AutoML pipelines for demand forecasting.
  - System design for ML-based segmentation of e-commerce customers.

Arute Solutions, Istanbul, Turkey

- Senior ML Scientist, Team Lead Sep 2017 – Jun 2019
  - Led a team of five ML scientists and engineers for the development and implementation of the flagship AI product: cash-demand forecasting deployed in production across 8,000+ ATMs.

EDUCATION

Boğaziçi University, Istanbul, Turkey

- PhD in Computer Engineering Sep 2016 – Sep 2023
  - Focus: Latent variable models, compressibility and generalization
  - Advisor: Prof. A. Taylan Cemgil
  - Cumulative GPA: 4.00 / 4.00
- B.A., M.A. in Psychology Sep 2007 – Aug 2014
  - Focus: Embodied language processing, causal explanations
  - Thesis: The effect of narrative shifts on perceptual simulations

SELECTED PUBLICATIONS AND MANUSCRIPTS

[17] **Barsbey, M.**, Prieto, L., Stevinson, E., & Birdal, T. Reliability under Pressure: How Quantized Representations Affect Safety, Alignment, and Post-Training Stability Through Layerwise Error Propagation. *In preparation*, 2026.

[16] **Barsbey, M.**, Ribeiro, A. H., Şimşekli, U., & Birdal, T. On the Interaction of Compressibility and Adversarial Robustness. *International Conference on Learning Representations (ICLR)*, 2026.

[15] **Barsbey, M.**, Stevinson, E., Prieto, L., & Birdal, T. Vulnerability to Multimodal Jailbreak and Prompt Injection Attacks under Activation Quantization. *ECCV Workshop on Universal Representations for Perception, Reasoning, and World Modeling*, 2026.

[14] Stevinson, E., Prieto, L., **Barsbey, M.**, & Birdal, T. Adversarial Vulnerability from Interference Between Features in Superposition. *International Conference on Machine Learning (ICML)*, 2026.

[13] Prieto, L., Stevinson, E., **Barsbey, M.**, Birdal, T., & Mediano, P. A. M. Correlations in the Data Lead to Semantically Rich Feature Geometry Under Superposition. *International Conference on Learning Representations (ICLR)*, 2026.

[12] **Barsbey, M.**, Prieto, L., Zafeiriou, S., & Birdal, T. Large Learning Rates Simultaneously Achieve Robustness to Spurious Correlations and Compressibility. *International Conference on Computer Vision (ICCV)*, 2025.

[11] Prieto, L., **Barsbey, M.**, Mediano, P. A. M., & Birdal, T. Grokking at the Edge of Numerical Stability. *International Conference on Learning Representations (ICLR)*, 2025.

[10] Stutz, D.*, Cemgil, A. T.*, Roy, A. G.*, Matejovicova, T.*, **Barsbey, M.***, et al. Evaluating AI Systems under Uncertain Ground Truth: A Case Study in Dermatology. *Medical Image Analysis, 103*, p.103556, 2025. *Equal contribution.

[9] Wan, Y.*, **Barsbey, M.***, Zaidi, A., & Şimşekli, U. Implicit Compressibility of Overparametrized Neural Networks Trained with Heavy-Tailed SGD. *International Conference on Machine Learning (ICML)*, 2024. *Equal contribution.

[8] Dvijotham, K.*, Winkens, J.*, **Barsbey, M.***, Ghaisas, S.*, Pawlowski, N. et al. Enhancing the reliability and accuracy of AI-enabled diagnosis via complementarity-driven deferral to clinicians (CoDoC). *Nature Medicine*, 29 (7), 1814-1820, 2023. *Equal contribution.

[7] **Barsbey, M.**, & Cemgil, A. T. Modeling Hierarchical Seasonality through Low-Rank Tensor Decompositions in Time Series Analysis. *IEEE Access*, 11, 85770-85784, 2023.

[6] **Barsbey, M.***, Özçelik, R.*, Bağ, A. et al. pydebiaseddta: A Computational Software for Training Robust Drug-Target Affinity Prediction Models. *Journal of Computational Biology*, 30 (11), 1240-1245, 2023. *Equal contribution.

[5] Raj, A., **Barsbey, M.**, Gürbüzbalaban, M., et al. Algorithmic Stability of Heavy-Tailed Stochastic Gradient Descent on Least Squares. *Algorithmic Learning Theory (ALT)*, 2023.

[4] **Barsbey, M.***, Sefidgaran, M.*, Erdogdu, M. A., Richard, G., & Şimşekli, U. Heavy Tails in SGD and Compressibility of Overparametrized Neural Networks. *Advances in Neural Information Processing Systems Conference (NeurIPS)*, 2021. *Equal contribution.

[3] Kurutmaz, M. B.*, **Barsbey, M.***, Cemgil, A. T., Yıldırım, S., & Şimşekli, U. Bayesian Model Selection for Identifying Markov Equivalent Causal Graphs. *Symposium on Advances in Approximate Bayesian Inference (AABI)*, 2019. *Equal contribution.

[2] **Barsbey, M.**, Sarıbay, S. A., Boduroğlu, A., & Zwaan, R. A. Perceptual simulations beyond the sentence: the effect of narrative shifts on perceptual simulations. *Annual Conference on Embodied and Situated Language Processing*, 2014.

[1] **Barsbey, M.**, & Lombrozo, T. Linguistic differences in explanation requests and their effects on the evaluation of explanations: the case of English and Turkish. *CogSci Conference*, 2013.

TECHNICAL SKILLS

Languages & frameworks	Python, Julia; PyTorch, JAX
Training & inference	PyTorch Lightning, HF Transformers, TRL, PEFT, vLLM
Engineering	Distributed training, SLURM, Linux, Docker, Git, pytest
Compression	bitsandbytes, torchao, torch-pruning; low-precision (FP8, NVFP4)
Experiment tooling	Hydra, Weights & Biases, MLflow

SCIENTIFIC SERVICE

Area Chair

- International Conference on Learning Representations (ICLR) 2026 – Present

Reviewer

- Neural Information Processing Systems (NeurIPS) 2019 – Present
- International Conference on Machine Learning (ICML) 2021 – Present
- International Conference on Learning Representations (ICLR) 2021 – Present
- Conference on Computer Vision and Pattern Recognition (CVPR) 2024 – Present
- AAAI Conference on Artificial Intelligence (AAAI) 2026 – Present

RECOGNITION & SCHOLARSHIPS

- UK Global Talent Visa, endorsed by UKRI 2024 – 2029
- PhD Scholarship from The Scientific and Technological Research Council of Turkey (TÜBİTAK) 2016 – 2021